

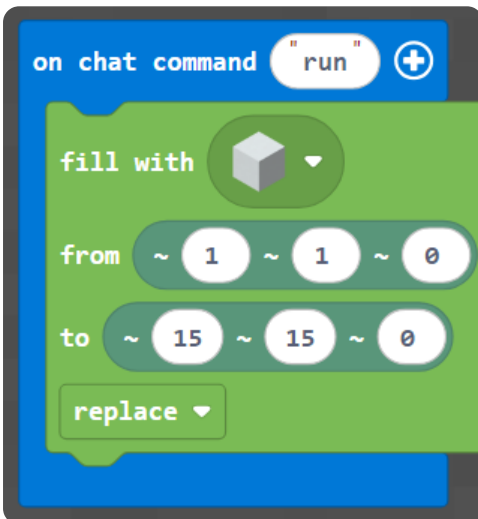
M003 · Pixel Art 2 — Cat & Tree (x, y) — English Worksheet (Intermediate)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate · **Time:** about 40 minutes

This lesson you copy a **tree** onto your wall. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

1 · Predict 🧠

Read each set of steps. Write a sentence saying what you think you will see.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
place brown block at (8, 4)
```

Across or up? Which number stays the same? (This is the tree's trunk.)


```
place orange block at (2, 15)
place orange block at (3, 15)
```

Across or up? Which number stays the same?


```
place green block at (8, 15)
place green block at (7, 14)
place green block at (9, 14)
```

These green blocks sit high and spread out going down. Is this a fruit, or the top of the tree?

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why it was wrong and your fix works.

Bug A — This should make the **tree trunk**: four brown blocks going **up** at $x = 8$, from $(8, 1)$ to $(8, 4)$. Right now they all land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

Hint: to move up, the **y** has to change each time.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The tree trunk should grow **up** at $x = 8$: $(8, 2)$, $(8, 3)$, $(8, 4)$. Right now it grows **across** instead.

```
place brown block at (2, 8)
place brown block at (3, 8)
place brown block at (4, 8)
```

Hint: check which slot the changing number is in — x (across) or y (up). The two numbers got swapped.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world and stand on your **home spot** (the red block). Copy the **tree** below onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

🌳 Tree

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15						■	■	■	■	■					
14					■	■	■	■	■	■	■				
13				■	■	■	■	■	■	■	■	■			
12			■	■	■	■	■	■	■	■	■	■	■		
11			■	■	■	■	■	■	■	■	■	■	■		
10		■	■	■	■	■	■	■	■	■	■	■	■	■	
9		■	■	■	■	■	■	■	■	■	■	■	■	■	
8		■	■	■	■	■	■	■	■	■	■	■	■	■	
7			■	■	■	■	■	■	■	■	■	■	■		
6				■	■	■	■	■	■	■	■	■			
5					■	■	■	■	■	■					
4						■	■	■							
3						■	■	■							
2						■	■	■							
1						■	■	■							

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.